

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	M.Rishitha	Reg. No.:	192424335
Section / Batch:	2024 AIDS	Date:	26-08-2026

Code of Conduct

I M.Rishitha[192424335] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.Rishitha

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- Gender and number agreement
- Recency
- Grammatical role
- Semantic compatibility
- Discourse coherence

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
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d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. Maintain entity coherence using assignment, mistakes, and you.
2. Use a Cause–Effect or Elaboration discourse relation.
3. Include at least two keywords from:
 - review
 - improve
 - confident
 - feedback
4. Response length must be 2–3 sentences.

5. Maintain a positive and polite tone.
6. Do not give contradictory advice.

Tasks:

- a) Identify the dialog acts required.
- b) Identify the important entities that must be maintained.
- c) Generate three possible responses satisfying all constraints.
- d) Compare the responses based on:
 - Coherence
 - Constraint satisfaction
 - Politeness
 - Relevance
- e) Select the best response and justify your answer.
- f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- A financial institution
- The side of a river

Constraints:

1. Resolve the correct word sense using contextual information.
2. Identify important entities and actions.
3. Represent the meaning using predicate logic.
4. Preserve semantic relationships.
5. Maintain coherence in the generated paraphrase.

Tasks:

- a) Identify the possible meanings of bank.
- b) Select the correct sense and justify your decision using contextual constraints.
- c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1	7	7	7	6	6	33	93
2	7	7	6	6	6	32	
3	6	6	6	5	5	28	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

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Question 1 — Reference Resolution and Discourse Coherence

Given paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

a) Referring expressions and their possible antecedents

- "He" (S2) → possible antecedents: Ravi, Arun
- "him" (S3) → possible antecedents: Ravi, Arun
- "it" (S3) → possible antecedent: "a book on Artificial Intelligence"
- "they" (S4) → possible antecedent: Ravi and Arun (combined plural referent)
- "the book" (S4) → possible antecedent: "a book on Artificial Intelligence" / "it"

b) Applying resolution constraints

- Gender and number agreement: All of Ravi, Arun and "he/him" are singular masculine, so this constraint alone cannot disambiguate between Ravi and Arun; however it correctly rules out the book (neuter) as a candidate for "he/him", and confirms "it" must refer to the (neuter, singular) book, while "they" must refer to a plural set → Ravi + Arun.
- Recency: The most recently mentioned entity before "He" is Arun ("Ravi met Arun"), so recency alone would weakly favour Arun.
- Grammatical role (subject preference): Ravi is the grammatical subject of the first sentence. Subsequent subject pronouns tend to co-refer with a preceding subject (parallel-function preference), so "He" is preferred to resolve to Ravi rather than the object Arun.
- Semantic compatibility: "looking for a book" and later "Arun helped him find it" are only sensible if the searcher (him) is different from the helper (Arun); this is compatible with Ravi as the searcher and Arun as the helper, and with "it" referring to the inanimate book.
- Discourse coherence: The narrative is coherent only if the person who came to the library looking for a book (He/him = Ravi) is helped by the other person (Arun), after which both (they) discuss the book — this rules out Arun as antecedent of "He/him" since it would make Arun help himself.

c) Resolution table

Referring Expression	Possible Antecedents	Valid / Invalid	Reason
He	Ravi, Arun	Ravi – Valid Arun – Invalid	Grammatical role (subject parallelism) and discourse coherence favour Ravi; recency alone would suggest Arun but is overridden.
him	Ravi, Arun	Ravi – Valid Arun – Invalid	Semantic compatibility: Arun is the subject who "helped", so the

Referring Expression	Possible Antecedents	Valid / Invalid	Reason
			object "him" must be the other participant, Ravi.
it	the book on AI	Valid	Gender/number agreement (neuter, singular) and semantic compatibility with an inanimate object being found.
they	Ravi and Arun	Valid	Number agreement (plural) and discourse coherence — both participants are present to discuss the book.
the book	a book on Artificial Intelligence / it	Valid	Same referent reintroduced by definite description; consistent with prior mentions.

d) Final coreference chains

- Chain 1 (Ravi): Ravi → He → him
- Chain 2 (Arun): Arun → Arun (explicit second mention)
- Chain 3 (Book): a book on Artificial Intelligence → it → the book
- Chain 4 (Ravi + Arun): they

e) Paragraph rewritten with resolved references

"Ravi met Arun at the library. Ravi was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving."

f) Ambiguity if "He was looking for a book" appeared before Arun is introduced

If the pronoun "He" occurred before Arun was mentioned at all, there would be no second candidate antecedent yet in the discourse, so a reader would have only one available referent (Ravi) and might provisionally, but not confidently, commit to it — this is a case of cataphora, where the pronoun is used before its antecedent is fully established. Once Arun is introduced afterward, the reader must retrospectively check whether "He" should instead be reassigned to Arun, since both are plausible. This forced re-evaluation increases processing load and genuine referential ambiguity, because the usual disambiguating constraints (recency, grammatical role parallelism) cannot be applied until the whole sentence, or later sentences, have been read. In short, reordering the pronoun before its antecedent removes the linear cues the resolution constraints depend on and creates temporary, and potentially lasting, ambiguity about who "He" refers to.

Question 2 — Dialog Act Recognition and Response Generation

Conversation history: Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

a) Dialog acts required

- Reassure — acknowledges the student's anxiety and reduces worry (e.g., normalising the feeling of doubt after submission).

- Advise — gives a concrete, constructive suggestion for what the student should do next (e.g., review the work, seek feedback).

b) Important entities to maintain

- assignment — the object of concern
- mistakes — the source of the student's worry
- you — the student, who is the target of reassurance and advice

c) Three candidate responses satisfying all constraints

- R1: "Don't worry — it's completely normal to feel unsure after submitting an assignment. I'd suggest you review your work and note what to improve, so you can feel more confident next time."
- R2: "It's okay if you're not fully sure about it — many students feel that way. Try to review the assignment for any mistakes and ask for feedback so you can improve going forward."
- R3: "You've already done the hard part by submitting it! A quick review afterward can help you catch small mistakes and build your confidence for the next one."

d) Comparison of the three responses

Response	Coherence	Constraint Satisfaction	Politeness	Relevance
R1	High – clear Cause→Effect (review → improve → confident)	Full – uses review, improve, confident; keeps assignment/mistakes/you; 2 sentences	High – warm, encouraging opening	High – directly answers the worry
R2	Good – Elaboration relation, slightly generic	Good – uses review, feedback, improve; 2 sentences	High – normalises the feeling	High – on-topic but less specific
R3	Good – positive reframing before advice (Elaboration)	Partial – uses review, confident but not improve/feedback explicitly	Very high – praises effort first	High – relevant, upbeat tone

e) Best response and justification

R1 is the best response. It satisfies every stated constraint simultaneously: it keeps all three entities in focus (assignment, mistakes implied through "review your work", and you), uses a clear Cause–Effect discourse relation (reviewing leads to improvement leads to confidence), includes three of the required keywords (review, improve, confident), stays within the 2–3 sentence limit, maintains a positive and polite tone throughout, and gives a single, non-contradictory piece of advice.

f) Effect of not maintaining entity coherence

If entity coherence is not maintained — for example, if the response suddenly talks about an unrelated exam or a different assignment, or switches who "you" refers to — the response becomes disjointed and harder to follow. The listener has to work out which entity is being discussed at each point, discourse relations like Cause–Effect break down because they depend on a shared thread of entities, the reply feels less relevant and less trustworthy, and it becomes easier to accidentally give advice that contradicts what was said earlier, since the response is no longer anchored to the same assignment, mistakes, and student throughout.

Question 3 — Word Sense Disambiguation and Meaning Representation

Source sentence:

"Priya sat on the bank and watched the boats moving across the water."

a) Possible meanings of "bank"

- Sense 1: A financial institution.
- Sense 2: The land alongside a river (a riverbank).

b) Correct sense and justification

The correct sense is Sense 2 — the side of a river. The surrounding context words "boats", "moving across", and "water" are all semantically associated with a riverside/waterside setting, not a financial institution. Selectional restrictions also support this: one can literally "sit on" a riverbank, but "sit on" a financial institution is not a physically plausible action. This contextual and selectional compatibility resolves the ambiguity in favour of the riverbank sense.

c) Predicate logic representation

$$\exists x \exists y \exists z \exists w [\text{Priya}(x) \wedge \text{bank}(y) \wedge \text{sit}(x) \wedge \text{beside}(x, y) \wedge \text{boat}(z) \wedge \text{watch}(x, z) \wedge \text{move}(z) \wedge \text{water}(w) \wedge \text{across}(z, w)]$$

Reading: there exists an x, y, z and w such that x is Priya, y is a (river)bank, x sits and is beside y, z is (a set of) boats, x watches z, z moves, w is water, and z moves across w.

d) Unambiguous English paraphrase

"Priya sat beside the riverbank and watched the boats crossing the water."

e) How Word Sense Disambiguation improves Machine Translation

- Correct lexical choice: WSD ensures the translator picks the target-language word matching the intended sense (e.g., a riverbank word rather than a financial-institution word), avoiding embarrassing mistranslations.
- Avoids semantic errors: many languages use entirely different words for different senses of an English polysemous word, so failing to disambiguate can make the translated sentence meaningless or nonsensical.
- Improves fluency and naturalness: choosing the sense-appropriate word keeps collocations natural in the target language (e.g., pairing "riverbank" with "boats" and "water").
- Preserves meaning and coherence: correct sense selection keeps the semantic relationships (who did what to whom) intact across languages, which is essential for accurate, coherent translation output.